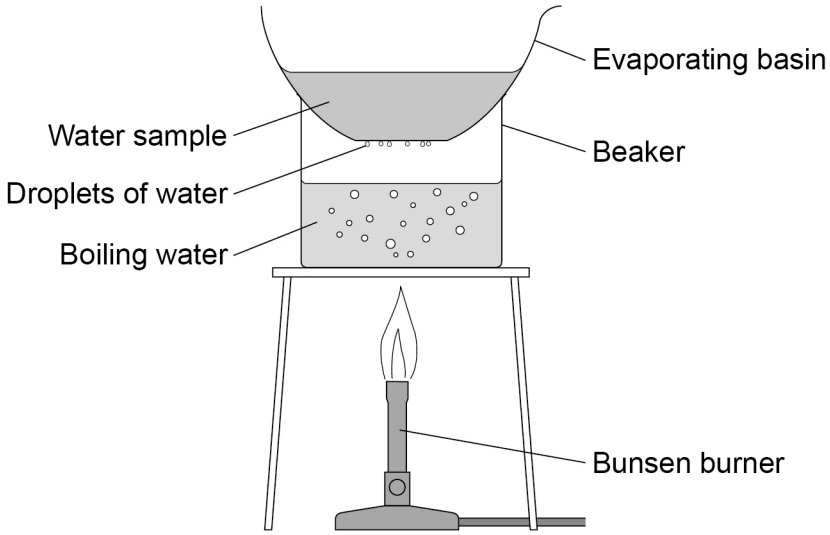
Do not write outside the box

0 3 A student investigated the mass of dissolved solids in water samples.

Figure 4 shows the apparatus used.

Figure 4



This is the method used.

1. Record the mass of a dry evaporating basin.
2. Pour 25 cm^3 of the water sample into the evaporating basin.
3. Place the evaporating basin on the beaker for 10 minutes.
4. Record the mass of the evaporating basin and contents.

0 3 . 1 What is used to find the mass of the evaporating basin?

[1 mark]

Tick (✓) **one** box.

Balance ☐

Beaker ☐

Measuring cylinder ☐

Thermometer ☐

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One error is that droplets of water collect on the bottom of the evaporating basin.

0 3 . 2 Suggest how this error affects the mass of the evaporating basin and contents.

[1 mark]

_____

_____

0 3 . 3 How can this error be corrected?

[1 mark]

_____

_____

0 3 . 4 Another error in the method is that not all the water was removed from the water sample.

How can this error be corrected?

[1 mark]

Tick (✓) **one** box.

Add more boiling water to the beaker. ☐

Heat until the mass of the evaporating basin and contents is constant. ☐

Stir the water sample in the evaporating basin with a glass rod. ☐

Question 3 continues on the next page

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0 3 . 5 The water in the water sample turns into steam.

What is the name of this process?

[1 mark]

Another student did the experiment correctly with three water samples **A**, **B** and **C**.

Table 2 shows the results.

Table 2

Water sample	Mass of dissolved solids in g			
	Test 1	**Test 2**	**Test 3**	**Mean**
A	0.23	0.23	0.20	**X**
B	0.03	0.07	0.02	0.04
C	1.45	1.60	1.45	1.50

0 3 . 6 The range is the difference between the largest value and the smallest value.

Which water sample has the greatest range of results?

[1 mark]

Tick (✓) **one** box.

A ☐

B ☐

C ☐

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0 3 . 7 Calculate the mean mass **X** for water sample **A**.

Use **Table 2**.

[2 marks]

_____

_____

_____

_____

X = _____ g

0 3 . 8 What is the dependent variable in this experiment?

[1 mark]

Tick (✓) **one** box.

Mass of dissolved solids ☐

Time taken for water to heat ☐

Type of water sample ☐

Volume of boiling water ☐

0 3 . 9 A different water sample contains 3.6 g of dissolved solids in 150 cm^3

Calculate the mass of dissolved solids in 25 cm^3 of this sample.

[2 marks]

_____

_____

_____

_____

Mass = _____ g

11

Turn over ►